

Formula Summary

3D Mapping

$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_e \\ t_e \\ t_e \end{bmatrix} + R_n^e(L, B) R_b^n(\phi, \theta, \psi) \cdot \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + R_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ b \cdot \sin(b) \\ b \cdot \cos b \end{bmatrix}$$

From Sensor actually, but:

- No rotation around the vertical axis ($a=0$)

-

$$\begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} = \begin{bmatrix} d \cdot \sin b \cdot \sin a \\ d \cdot \sin b \cdot \cos a \\ d \cdot \cos b \end{bmatrix} = \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$


$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$

Graph SLAM

DLT: Direct Linear Transformation - Affine Transformation-

- Mapping $\chi = \mathcal{P}(\chi) : \mathbf{x} = P\mathbf{X}$

- with $P = KR[I_3] - \mathbf{X}_O$

- and $K = \begin{bmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{bmatrix}$

- contains 5 intrinsic parameters:

- Camera constant c

- Principal point: x_H, y_H

- Scale difference: m

- Shear parameter: s - typically for digital cameras $s = 0$

Accounting for non-linear Errors

$${}^a\mathbf{x} = {}^aH_s({}^s\mathbf{x}) {}^s\mathbf{x}$$

$$\text{with } {}^aH_s({}^s\mathbf{x}) = \begin{bmatrix} 1 & 0 & \Delta x(\mathbf{x}, \mathbf{q}) \\ 0 & 1 & \Delta y(\mathbf{x}, \mathbf{q}) \\ 0 & 0 & 1 \end{bmatrix}$$

→ overall mapping is:

$${}^a\mathbf{x} = {}^aH_s({}^s\mathbf{x}) KR[I_3 | -\mathbf{X}_O]\mathbf{X}$$

DLT from Point-Pixel Correspondences

- P contains **11 parameters**
 - 6 extrinsic parameters: $R, \mathbf{X}_O$
 - 5 intrinsic parameters: c, x_H, y_H, m, s
 - Necessary: ≥ 6 points (Each point **2** observation equations - one for each image coordinate)

1. Build the M for the linear system

$$M = \begin{bmatrix} \mathbf{a}_{x_1}^\top \\ \mathbf{a}_{y_1}^\top \\ \vdots \\ \mathbf{a}_{x_I}^\top \\ \mathbf{a}_{y_I}^\top \end{bmatrix} \quad \begin{array}{c} M \mathbf{p} \stackrel{!}{=} 0 \\ \nearrow \quad \nwarrow \\ 2I \times 12 \quad 12 \times 1 \end{array}$$

with

$$\begin{aligned} \mathbf{a}_{x_i}^\top &= (-X_i, -Y_i, -Z_i, -1, 0, 0, 0, 0, x_i X_i, x_i Y_i, x_i Z_i, x_i) \\ \mathbf{a}_{y_i}^\top &= (0, 0, 0, 0, -X_i, -Y_i, -Z_i, -1, y_i X_i, y_i Y_i, y_i Z_i, y_i) \end{aligned}$$

DLT in a Nutshell

2. Solve by SVD $M = U S V^\top$ Solution is last column of V

$$\mathbf{p} = \mathbf{v}_{12} \Rightarrow P = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix}$$

DLT in a Nutshell

3. If individual parameters are needed

$$P = KR [I_3 | -X_O] = [H|h]$$

$$X_O = -H^{-1} h$$

$$QR(H^{-1}) = R^T K^{-1}$$

$$R \leftarrow R(z, \pi) R$$

$$K \leftarrow \frac{1}{K_{33}} KR(z, \pi)$$

- For negative camer constant use:

$$R(z, \pi) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **NO solution**, if all points X_i are located on a **plane** -> Rank deficiency

Photogrammetric Model

- Object reconstructable only up to **similarity transform**
 - Orientation of Photogrammetric model
 - Absolute orientation (min. 3 point in 3D)

Cameras	# params per image	#params per img pair	# params for Relative Orientation	#params for absolute Orientation	minimum number of control points
Calibrated	6	12	5	7	3
Uncalibrated	11	22	7	15	5

- Computable: Rotation R and Direction B
- Use Coplanarity Constraint
- Fundamental Matrix: Straight-Line Preserving Mapping (Uncalibrated Cameras)
- Essential Matrix: Angle Preserving Mapping

Fundamental Matrix

$$\underbrace{\mathbf{x}'^T (\mathbf{K}')^{-T} (\mathbf{R}')^{-T} \mathbf{S}_b (\mathbf{R}'')^{-1} (\mathbf{K}'')^{-1} \mathbf{x}''}_F = 0$$
$$\mathbf{x}'^T F \mathbf{x}'' = 0$$

From Camera Projection Matrices

- We have
 - $P' = [A' | \mathbf{a}'] = \underbrace{[K' R']}_{A'} \underbrace{[-K' R' \mathbf{X}_{O'}]}_{\mathbf{a}'}$
- Recover Projection center
 - $\mathbf{X}_{O'} = \mathbf{A}'^{-1} \mathbf{a}'$
- Base line:
 - $\mathbf{b}'_{12} = \mathbf{A}''^{-1} \mathbf{a}'' - \mathbf{A}'^{-1} \mathbf{a}'$
- We can compute F:
 - $F = A'^{-T} S_{b'_{12}} A''^{-1}$
 - $S_b = \begin{bmatrix} 0 & -b_3 & b_2 \\ b_3 & 0 & -b_1 \\ -b_2 & b_1 & 0 \end{bmatrix}$

From corresponding points in Images

```
1 function F = F_from_point_pairs(xs, xss)
2 % xs, xss: Nx3 homologous point coordinates, N > 7
3 % F:      3x3 fundamental matrix
4
5 % coefficient matrix
6 for n = 1 : size(xs, 1)
7     A(n, :) = kron(xss(n, :), xs(n, :));
8 end
9
10 % singular value decomposition
11 [U, D, V] = svd(A);
12
13 % approximate F, possibly regular
14 Fa = reshape(V(:, 9), 3, 3);
15
16 % svd decomposition of F
17 [Ua, Da, Va] = svd(Fa);
18
19 % algebraically best F, singular
20 F = Ua * diag([Da(1, 1), Da(2, 2), 0]) * Va';
```

- Ensure Well-conditioned Problem: **Conditioning/Normalisation**
 - **Transformation:** center of mass of all points is at (0, 0)

- **Scale** image: $x, y \in [-1, 1]$

Essential Matrix

$$-{}^k\mathbf{x}'^T \mathbf{R}' \mathbf{S}_b \mathbf{R}''^T {}^k\mathbf{x}'' = 0$$

$$\mathbf{E}$$

$${}^k\mathbf{x}'^T \mathbf{E} {}^k\mathbf{x}'' = 0$$

- Essential Matrix is homogeneous & singular

8-Point Solution

```

1 function E = E_from_point_pairs(xs, xss)
2 % xs, xss: Nx3 homologous point coordinates, N > 7
3 % E:      3x3 essential matrix
4
5 % coefficient matrix
6 for n = 1 : size(xs, 1)
7     A(n, :) = kron(xss(n, :), xs(n, :));
8 end
9
10 % singular value decomposition
11 [U, D, V] = svd(A);
12
13 % approximate E, possibly regular
14 Ea = reshape(V(:, 9), 3, 3);
15
16 % svd decomposition of E
17 [Ua, Da, Va] = svd(Ea);
18
19 % algebraically best E, singular, satisfies
20 E = Ua * diag([1, 1, 0]) * Va';

```

build matrix A

solve Ae=0

build matrix Ea

compute SVD of Ea

**build matrix E from Ea
by imposing constraints**

Orientation Parameters from Essential Matrix

- Algebraic Solution (Hartley & Zissermann)
- Define

- Skew-Sym. Matrix: $Z = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- Rotation matrix: $W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

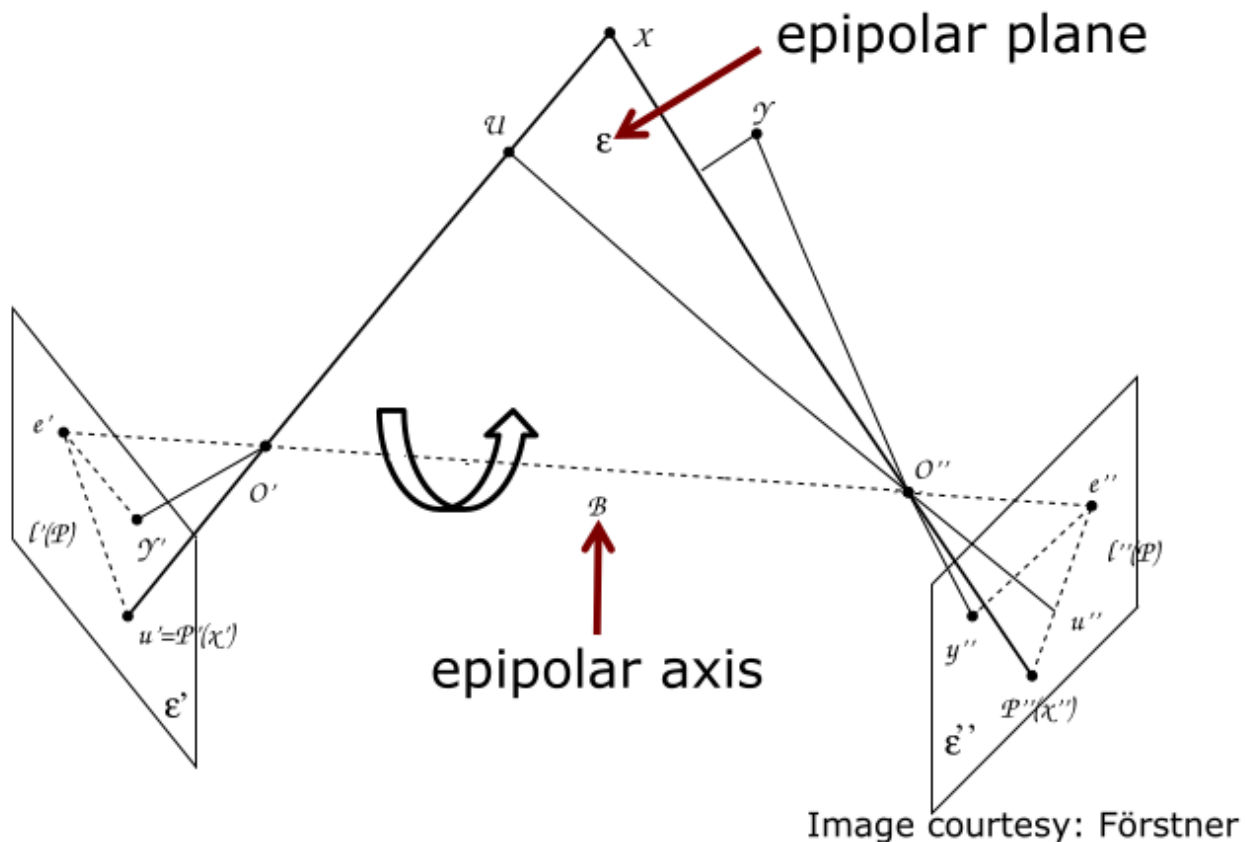
- So that $ZW = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- Compute the SVD of E: $UDV^T = \text{svd}(E)$
- Normalise U, V by $U = U|U|, V = V|V|$
- Compute 4 solutions:

- $S_{\hat{B}}^1 = UZU^T$
- $S_{\hat{B}}^2 = UZ^TU^T$
- $R_1^T = UWV^T$
- $R_2^T = UW^TV^T$

- Test for which solution all points are in front of both cameras

Epipolar Geometry



- **Epipolar Axis:** From projection centers
 - Homogenous vector $b = X_{O''} - X_{O'}$
 - **only direction, not length**
- **Epipolar Lines:** Lie on epipolar lines
 - Exploit coplanarity constraint for both points
 - $x'^T \frac{Fx''}{l'} = 0$
 - $\frac{x'^T Fx''}{l'^T} = 0$

◦ $\Rightarrow \boxed{l' = Fx''} \quad \boxed{l'' = F^T x'}$ **Prediction (1D space)**

- **Epipoles:** Projection Centers in other image
 - From projection Matrices

- $e' = P' X_{O''} \quad e'' = P'' X_{O'}$

- Is the Intersection of ALL Epipolar Lines in an Image

- $\forall l' : e'^T l' = 0 \quad \forall l'' : e''^T l'' = 0$

- $e'^T F x'' = 0$

- Epipoles is null space of F^T

- $x''^T F'^T e' = 0$

- Epipoles are left and right eigenvectors of the fundamental matrix

$$\begin{aligned} e'^T F &= 0 & F e'' &= 0 \\ \text{null}(F^T) &= e' & \text{null}(F) &= e'' \end{aligned}$$

Stereo Triangulation

- Given relative orientation of 2 images, compute points in 3D

- 2 lines in 3D

- $f = q + \lambda r$ and $g = p + \mu s$

- Solve

$$\begin{aligned} \begin{bmatrix} r^T r & -s^T r \\ r^T s & -s^T s \end{bmatrix} \begin{bmatrix} \lambda \\ \mu \end{bmatrix} &= \begin{bmatrix} (X_{O''} - X_{O'})^T r \\ (X_{O''} - X_{O'})^T s \end{bmatrix} \\ Ax &= b \end{aligned}$$

- H as the middle of connecting point F, G (retrieved from λ, μ)

$$H = \frac{F + G}{2}$$

Stereo Normal Case

1. Z-coordinate from intercept theorem

$$Z = c \frac{B}{-(x'' - x')}$$

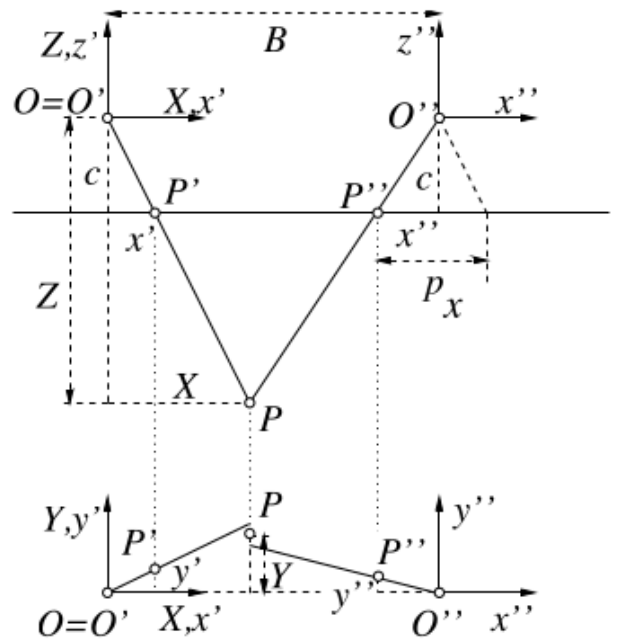
2. X-coordinate

$$X = x' \frac{B}{-(x'' - x')}$$

3. Y-coordinate by mean

$$\frac{Y}{X} = \frac{\frac{y' + y''}{2}}{x'}$$

$$\Rightarrow Y = \frac{y' + y''}{2} \frac{B}{-(x'' - x')}$$



- x-parallax $p_x = x'' - x'$ corresponds to depth Z
- y-parallax $p_y = y'' - y'$
 - corresponds to **consistency** of image points
 - Should be small (Stereo normal case)

$$\begin{bmatrix} U \\ V \\ W \\ T \end{bmatrix} = \underbrace{\begin{bmatrix} B & 0 & 0 & 0 \\ 0 & B & 0 & 0 \\ 0 & 0 & Bc & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}}_{\text{camera-dependent}} \underbrace{\begin{bmatrix} x' \\ y' \\ 1 \\ p_x \end{bmatrix}}_{\text{point-dependent}}$$

Absolute Orientation

- 2-Step Solution: Relative orientation, Absolute Orientation
- Similarity transform maps photogrammetric model m into the object reference frame o

$${}^o\mathbf{X}_n = \lambda \mathbf{R}^m \mathbf{X}_n + \mathbf{T}$$

- 7 DoF for similarity transform
 - (3 rotation, 3 translation, 1 scale)
- **Control points** are required

Least Squares Solution in Case of more control points

- Find similarity transform

$$\bar{\mathbf{x}}_n = \lambda \mathbf{R} \mathbf{x}_n + \mathbf{t}$$

- Minimise the squared error

$$\sum ||\mathbf{y}_n - \bar{\mathbf{x}}_n||^2 \rightarrow \min$$

Bundle Adjustment

Key Idea

- Start with an initial guess
- Project the **estimated 3D points** into the **estimated camera images**
- Compare locations of the **projected image points** with **measured ones**
- Adjust to minimize error in images

$${}^a\mathbf{x}'_{ij} + {}^a\hat{\mathbf{v}}_{x'_{ij}} = \hat{\lambda}_{ij} {}^a\hat{\mathbf{P}}_j(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i$$

a = "arbitrary frame"
 ij = point i observed in image j
 $\hat{\mathbf{v}}_{x'_{ij}}$ = corrections
 $\hat{\lambda}_{ij}$ = scale factor
 ${}^a\hat{\mathbf{P}}_j(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q})$ = projection matrix (w/ non-lin. calib.)
 $\hat{\mathbf{X}}_i$ = 3D point
 $\mathbf{x}_{ij}$ = img pix.
 $\mathbf{p}$ = project. params
 $\mathbf{q}$ = distortion

with $\sum x_{ij} x_{ij}$ $i = 1, \dots, I_j; j = 1, \dots, J$
 $\uparrow$ uncertainty in the image coordinates $\uparrow$ #points in image j $\uparrow$ #images

- Into Euclidian Space and avoiding $\hat{\lambda}_{ij}$

■ To
$${}^a\mathbf{x}'_{ij} + {}^a\hat{\mathbf{v}}_{x'_{ij}} = \frac{{}^a\hat{\mathbf{P}}_{1:2j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}{{}^a\hat{\mathbf{P}}_{3j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}$$

- Or written as an error term:

$$e(\mathbf{p}, \mathbf{q}, \hat{\mathbf{X}}_i) = {}^a\mathbf{x}'_{ij} - \frac{{}^a\hat{\mathbf{P}}_{1:2j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}{{}^a\hat{\mathbf{P}}_{3j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}$$

- For NL-LSA
 - $\hat{\Delta x} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} \Delta l$

- Linear System become huge

Structure

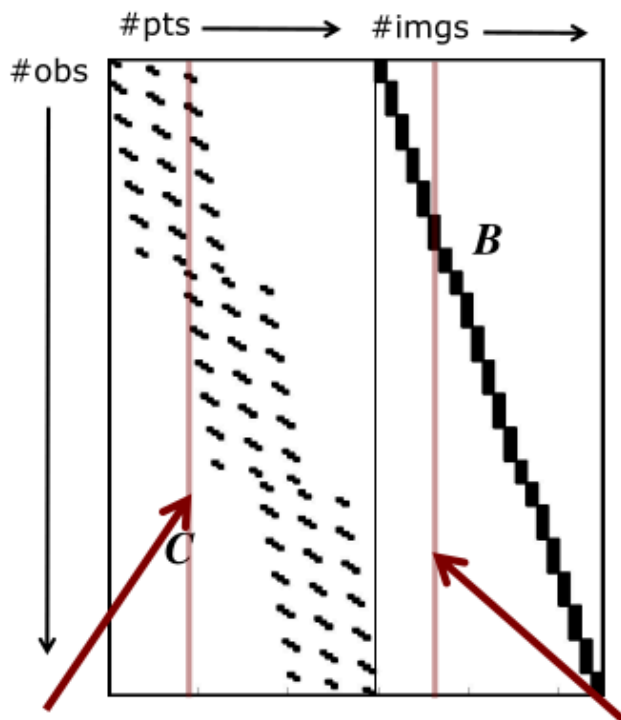
$$\Delta l + v = A \Delta x = \underbrace{\begin{bmatrix} C & B \end{bmatrix}}_A \underbrace{\begin{bmatrix} \Delta k \\ \Delta t \end{bmatrix}}_{\Delta x} = C \Delta k + B \Delta t$$

- Coefficient Matrix

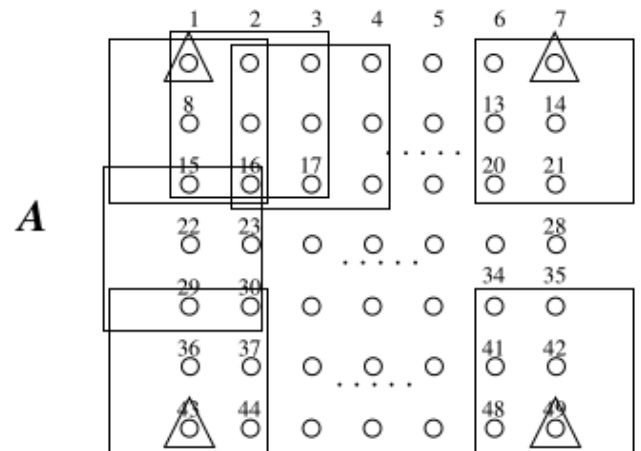
$$A = \begin{bmatrix} A_{2,1}^T \\ \vdots \\ A_{ij}^T \\ \vdots \\ A_{48,21}^T \end{bmatrix}$$

$$\text{with } A_{ij}^T = [0, \dots, 0, \underbrace{C_{ij}^T}_{2 \times 3}, 0, \dots, 0 | 0, \dots, 0, \underbrace{B_{ij}^T}_{2 \times 6}, 0, \dots, 0]$$

- -> **Sparse** A matrix



For 3 stripes with 7 images per stripe



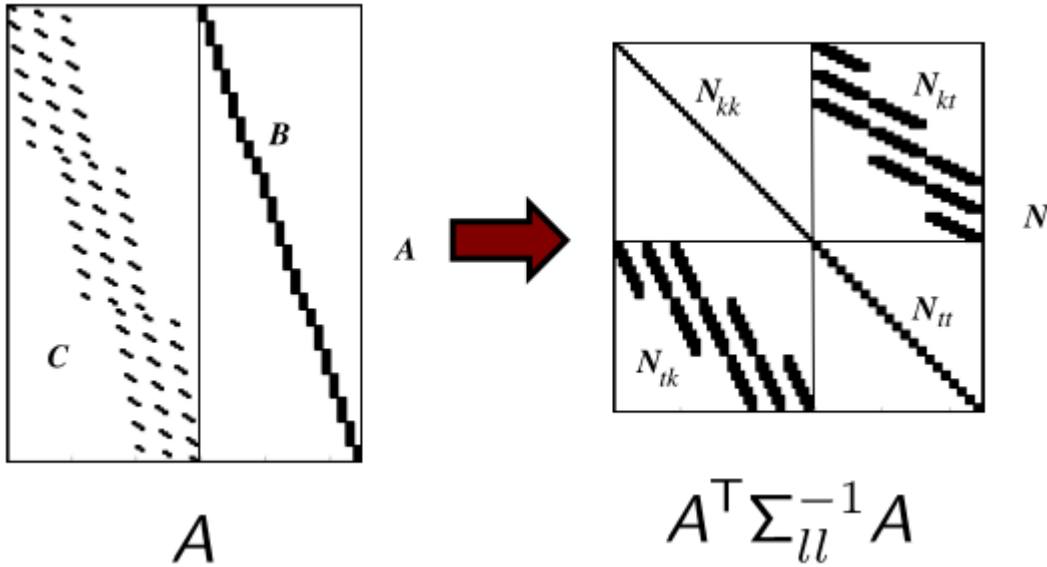
number of times a point is observed

number of observations (points) per image

- Properties of C:
 - 2×3 sub-matrices C_{ij}^T
 - Each connects an image point x'_{ij} with X_i
 - one per row
 - number of non-zero 2×3 matrices per column:
 - $\#X_i$ observed
- Properties of B:

- 2×6 sub-matrices B_{ij}^T
- Each connects an image point x'_{ij} with j^{th} camera orientation
- one per row
- number of non-zero 2×6 matrices per column:
 - $\#X_i$ observed

Covariance Matrix



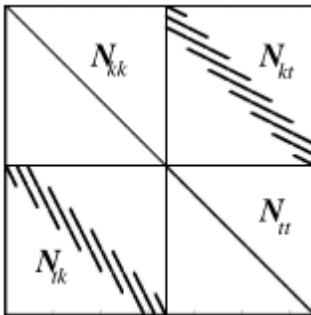
$$\begin{aligned}
 N &= A^T \Sigma_l^{-1} A = \begin{bmatrix} C^T \\ B^T \end{bmatrix} \Sigma_l^{-1} [C, B] \\
 &= \begin{bmatrix} C^T \Sigma_l^{-1} C & C^T \Sigma_l^{-1} B \\ B^T \Sigma_l^{-1} C & B^T \Sigma_l^{-1} B \end{bmatrix} = \begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix}
 \end{aligned}$$

block-diagonal
with 3x3 blocks

sparse matrix that
reveals the connections
of images and points;
consist of 3x6 blocks

$$N = \begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix}$$

Example:



block-diagonal
with 6x6 blocks

Keypoints

- Finding Corners
 - Computation of Sum of Squared Differences

$$f(x, y) = \sum_{(u,v) \in W_{xy}} (I(u, v) - I(u + \delta u, v + \delta v))^2$$

**local patch
around (x,y)**

**sum of squared differences
of image intensity values of
pixels under a given shift
(du, dv)**

◦

$$\begin{aligned} J_x^2 &= (D_x * I)^2 \\ J_x J_y &= (D_x * I)(D_y * I) \\ J_y^2 &= (D_y * I)^2 \end{aligned}$$

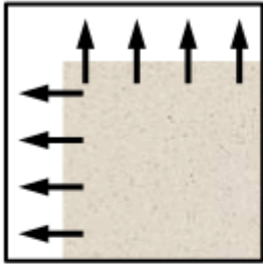
- Integrated and in Matrix form

$$f(x, y) \approx \sum_{(u,v) \in W_{xy}} \begin{bmatrix} \delta u & \delta v \end{bmatrix} \begin{bmatrix} J_x^2 & J_x J_y \\ J_x J_y & J_y^2 \end{bmatrix} \begin{bmatrix} \delta u \\ \delta v \end{bmatrix}$$

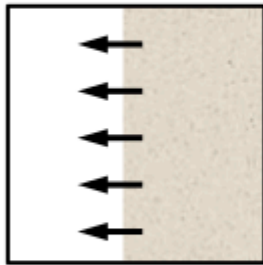
$$f(x, y) \approx [\delta u \quad \delta v] \begin{bmatrix} \sum_W J_x^2 & \sum_W J_x J_y \\ \sum_W J_x J_y & \sum_W J_y^2 \end{bmatrix} \begin{bmatrix} \delta u \\ \delta v \end{bmatrix}$$

- Using Taylor expansion, we obtain

$$I(u + \delta u, v + \delta v) \approx I(u, v) + [J_x \quad J_y] \begin{bmatrix} \delta u \\ \delta v \end{bmatrix}$$



$$\Rightarrow M = \begin{bmatrix} \gg 1 & \sim 0 \\ \sim 0 & \gg 1 \end{bmatrix} \quad \text{YES!}$$



$$\Rightarrow M = \begin{bmatrix} \sim 0 & \sim 0 \\ \sim 0 & \gg 1 \end{bmatrix} \quad \text{NO!}$$



$$\Rightarrow M = \begin{bmatrix} \sim 0 & \sim 0 \\ \sim 0 & \sim 0 \end{bmatrix} \quad \text{NO!}$$

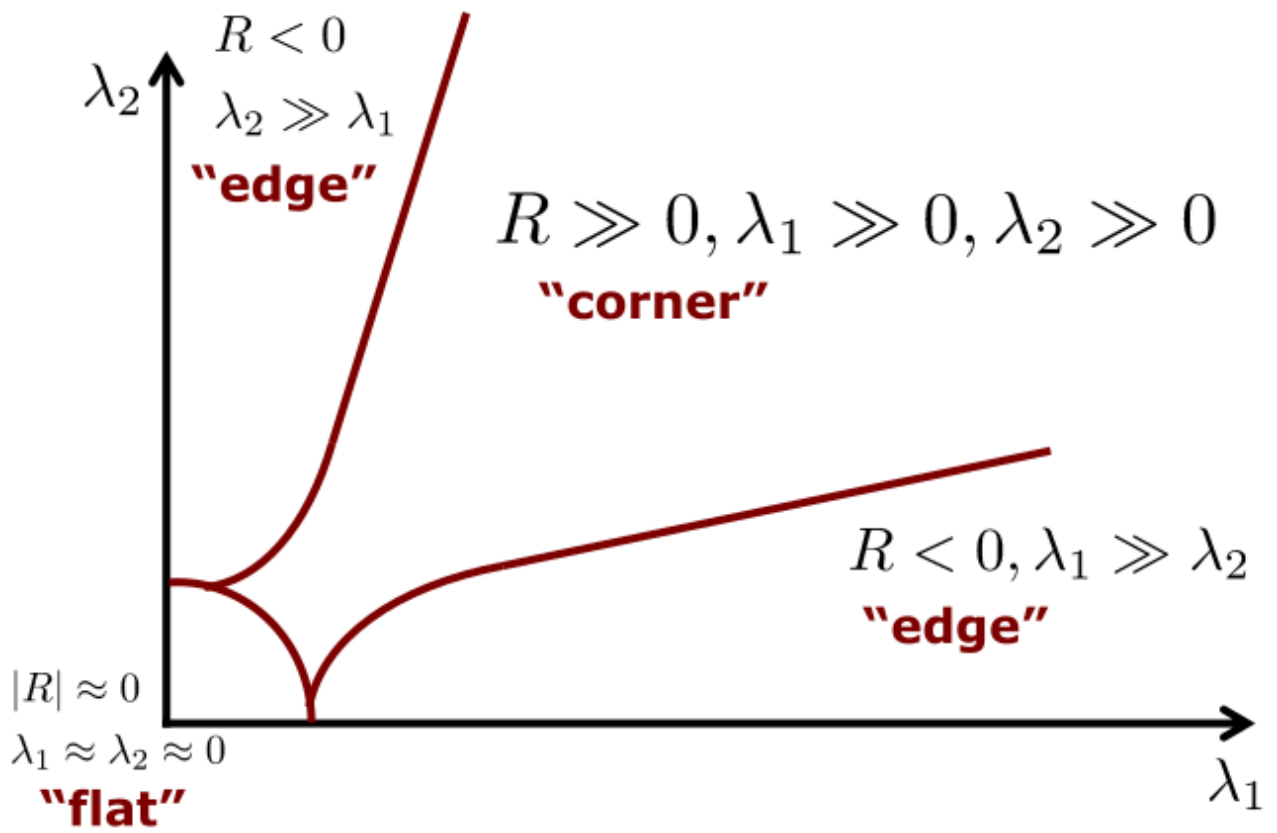
- Key Idea: 2 Large Eigenvalues
- Harris Corner Criterion

◦

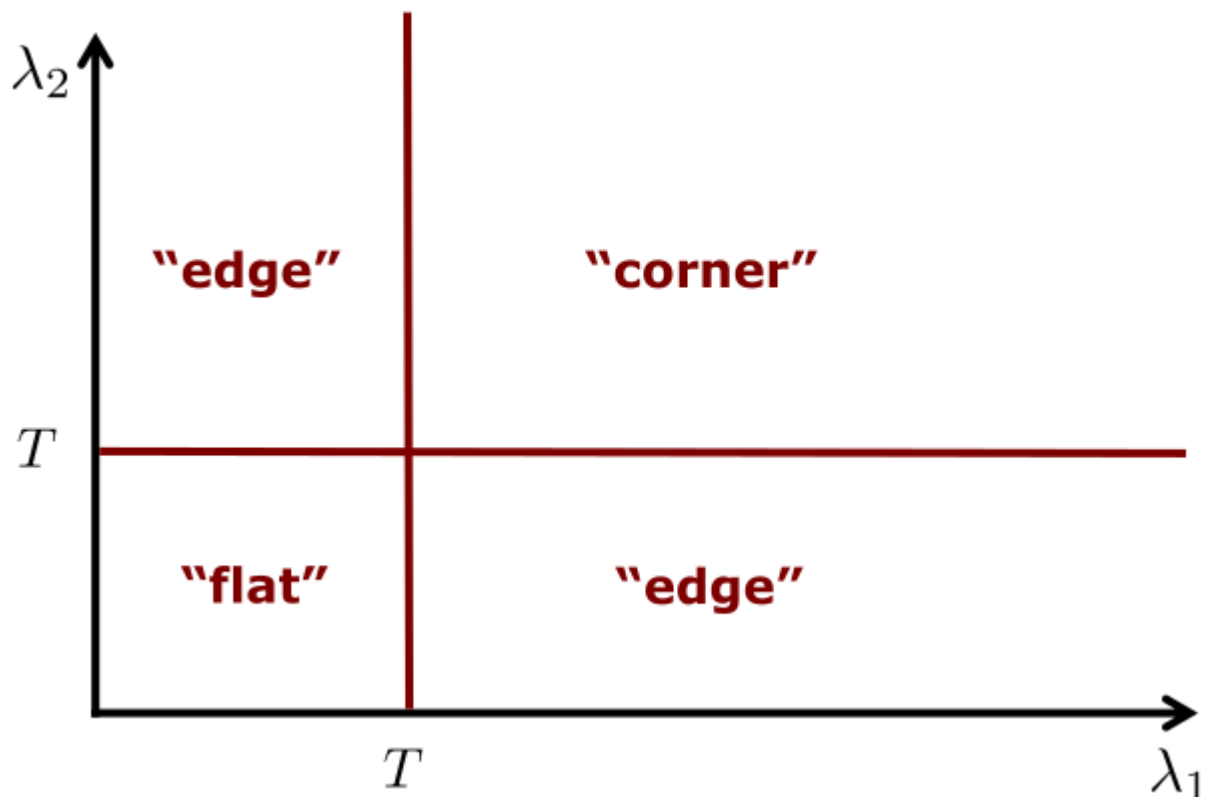
$$R = \det(M) - k(\text{trace}(M))^2$$

$$\blacksquare = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

- with $k \in [0.04, 0.06]$



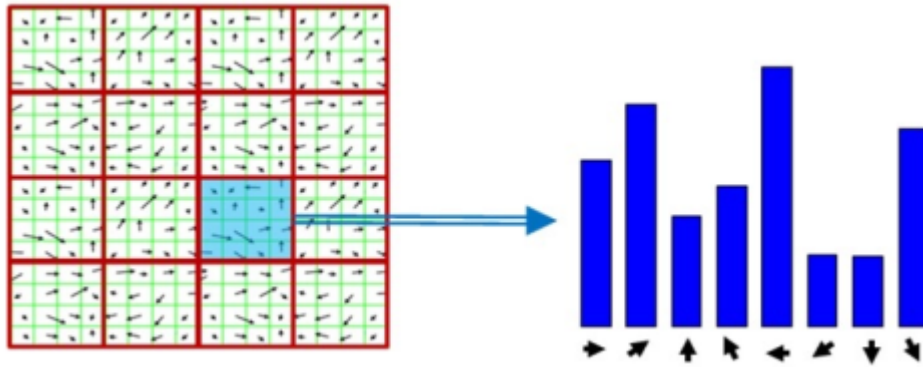
- Förstner Operator Criterion
 - Similar, but with inverse of M
 - size and roundness of error ellipse of covariance matrix
- Shi-Tomasi Corner Detector
 - $\lambda_{\min}(M) = \frac{\text{trace}(M)}{2} - \frac{1}{2}\sqrt{(\text{trace}(M))^2 - 4\det(M)}$
 - $\lambda_{\min}(M) \geq T$: corner



-
- Difference of Gaussians
 1. Gaussian smoothing
 2. Difference-of-Gaussians: find extreme (over smoothing scales)
 3. Maxima Suppression at edges

Feature Descriptors

- **SIFT: Scale Invariant Feature Transform**
 - Invariant to:
 - Image translation
 - rotation
 - scale
 - Suitable for detecting visual landmarks
 - from different angles and distances
 - with a different illumination
 - SIFT considers distribution of gradients around keypoints
 - local 16×16 area at selected scale
 - Divide pixels into 16, 4×4 pixel squares
 - Compute gradient direction histogram over 8 directions



-
- Concatenate histograms (128 feature vector)
- Match on Descriptor Difference
- **BRIEF: Binary Robust Independent Elementary**
 - Binary Descriptor:
 - Select patch around a keypoint
 - Select set of pixel pairs in that patch
 - For each pair, compare intensities
 - $$b = \begin{cases} 1 & \text{if } I(s_1) < I(s_2) \\ 0 & \text{otherwise} \end{cases}$$
 - Concatenate all b's to a bit string
 - For comparison: Use same pairs & maintain order of testing
- BRIEF: 5 different geometries (Uniform Random Sampling, Gaussian sampling...)
- **ORB: Oriented FAST and rotated BRIEF**
 - Extension to BRIEF:
 - Adds rotation compensation
 - Learns optimal sampling pairs
 - ORB defines strategy for
 - selecting 256 pairs
 - optimising for both properties using training database
- ORB: Faster, but not scale-invariant
- Mainly in-plan rotation invariant

ICP

- Find the parameters $R, \mathbf{t}$ of the rigid body transform with
 - $\bar{\mathbf{x}}_n = R\mathbf{x}_n + \mathbf{t} \quad n = 1, \dots, N$
- so that the squared error is minimised

$$\circ \sum ||\mathbf{y}_n - \bar{\mathbf{x}}_n||^2 \rightarrow \min$$

Direct Computation from known correspondences

1. Compute means of the point sets:

$$\mathbf{x}_0 = \frac{\sum \mathbf{x}_n p_n}{\sum p_n} \quad \mathbf{y}_0 = \frac{\sum \mathbf{y}_n p_n}{\sum p_n}$$

2. Compute cross Covariance matrix based on mean-reduced coordinated

$$H = \sum (\mathbf{x}_n - \mathbf{x}_0)(\mathbf{y}_n - \mathbf{y}_0)^T p_n$$

3. Compose H using SVD

$$svd(H) = UDV^T$$

4. Compute rotation

$$R = VU^T$$

5. Translation Vector

$$\mathbf{t} = \mathbf{y}_0 - R\mathbf{x}_0$$

PC Registration w/ unknown Data Associations

$$\bar{\mathbf{x}}_n = \mathbf{x}_n$$

error $e = \infty$

while (e has decreased and $e > \text{threshold}$)

$\mathcal{C} = \text{determine_correspondences}(\{\mathbf{y}_n\}, \{\bar{\mathbf{x}}_n\})$

$(\mathbf{t}, R) = \text{compute_transformation_params}(\mathcal{C}, \{\mathbf{y}_n\}, \{\bar{\mathbf{x}}_n\})$

$$\bar{\mathbf{x}}_n = R(\bar{\mathbf{x}}_n - \bar{\mathbf{x}}_0) + \mathbf{y}_0$$

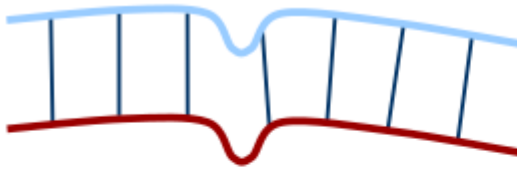
$$e = E(\mathbf{t}, R) = \Phi(R^\top \mathbf{y}_0 - R^\top \mathbf{t}, R)$$

return $\{\bar{\mathbf{x}}_n\}$

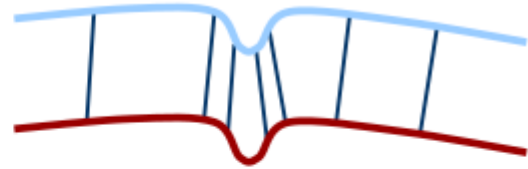
ICP Variants

1. Consider Point Subsets

- All points
- Random sampling
- Normal-space sampling
 - Good for mostly smooth areas with sparse features
 - Uniform sub-sampling



uniform sampling



normal-space sampling

- Feature-based sampling

2. Different data association strategies

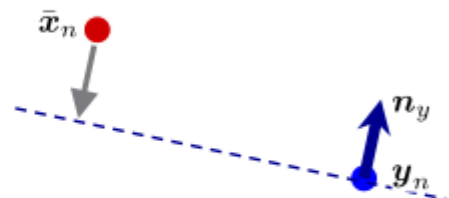
- Key for good results
- Closest point
- Closest compatible point
 - Normals, colors, curvature
- Normal shootin
- Point to Plane

point-to-point



$$\min \sum \| \mathbf{y}_n - \bar{\mathbf{x}}_n \|^2$$

point-to-plane



$$\min \sum ((\mathbf{y}_n - \bar{\mathbf{x}}_n) \cdot \mathbf{n}_y)^2$$

- Projection based approach

3. Weight the correspondences

$$\mathbf{y}_0 = \frac{\sum \mathbf{y}_n p_n}{\sum p_n} \quad \mathbf{x}_0 = \frac{\sum \mathbf{x}_n p_n}{\sum p_n}$$

$$H = \sum (\mathbf{x}_n - \mathbf{x}_0)(\mathbf{y}_n - \mathbf{y}_0)^\top p_n$$

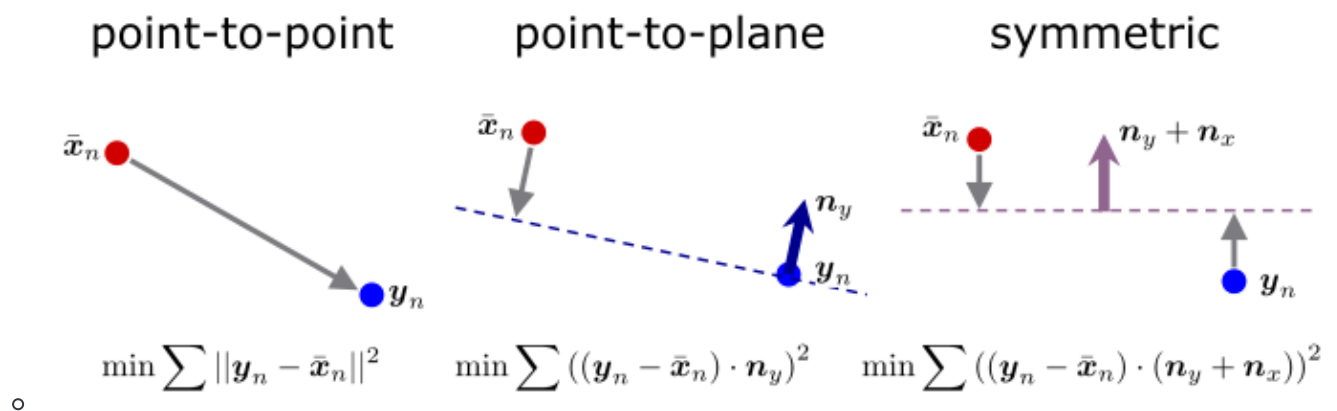
4. Reject potential outlier point pairs

- P2P distance larger than a given threshold

- Pairs not consistent with neighboring pairs
- Trimmed ICP: Sort correspondences w.r.t. their error, ignore worst t %

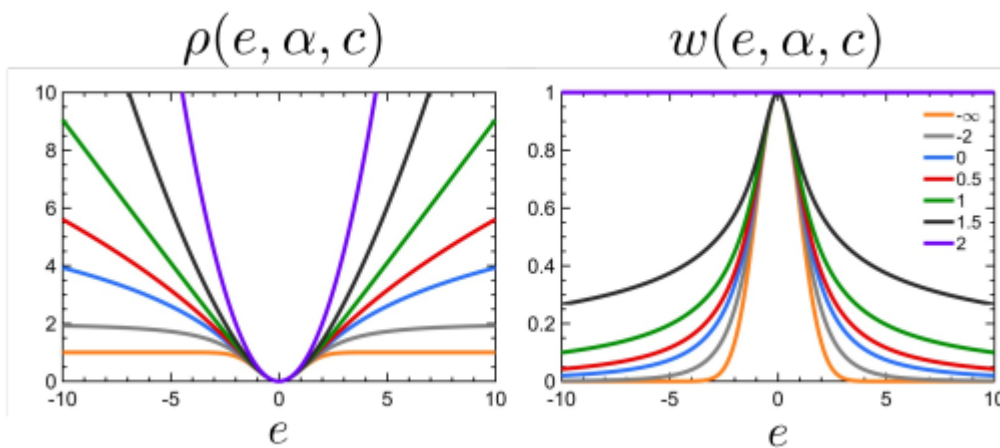
Robust Least Squares Approach

- Objective: $\Phi(t_x, t_y, \theta) = \sum \|\bar{\mathbf{x}}_n - \bar{\mathbf{y}}_n\| \rightarrow \min$
- Error Vector: $\mathbf{e}_n = \bar{\mathbf{x}}_n - \mathbf{y}_n = R\mathbf{x}_n + \mathbf{t} - \mathbf{y}_n$
- Parameters: $[t_x, t_y, \theta]^T$
- Linearise the non-linear error function
- Jacobian:
 - $J_n(\mathbf{x}) = \frac{\partial \mathbf{e}_n}{\partial \mathbf{x}}$
 - $J_n = \begin{bmatrix} 1 & 0 & -\sin \theta x_n - \cos \theta y_n \\ 0 & 1 & \cos \theta x_n - \sin \theta y_n \end{bmatrix}$
- Matrix: $H_n = J_n^T J_n$
- Right hand side: $\mathbf{b}_n = J_n^T \mathbf{e}_n$
- Compute terms over all points
 - $H = \sum_n H_n = \sum_n J_n^T J_n$
 - $\mathbf{b} = \sum_n \mathbf{b}_n = \sum_n J_n^T \mathbf{e}_n$
- Solve $H\Delta\mathbf{x} = -\mathbf{b}$ via $\Delta\mathbf{x} = -H^{-1}\mathbf{b}$
- Update parameters $\mathbf{x} \leftarrow \mathbf{x} + \Delta\mathbf{x}$
- Iterative until convergence
- Alternative Po2Pl symmetric



Robust Least Squares

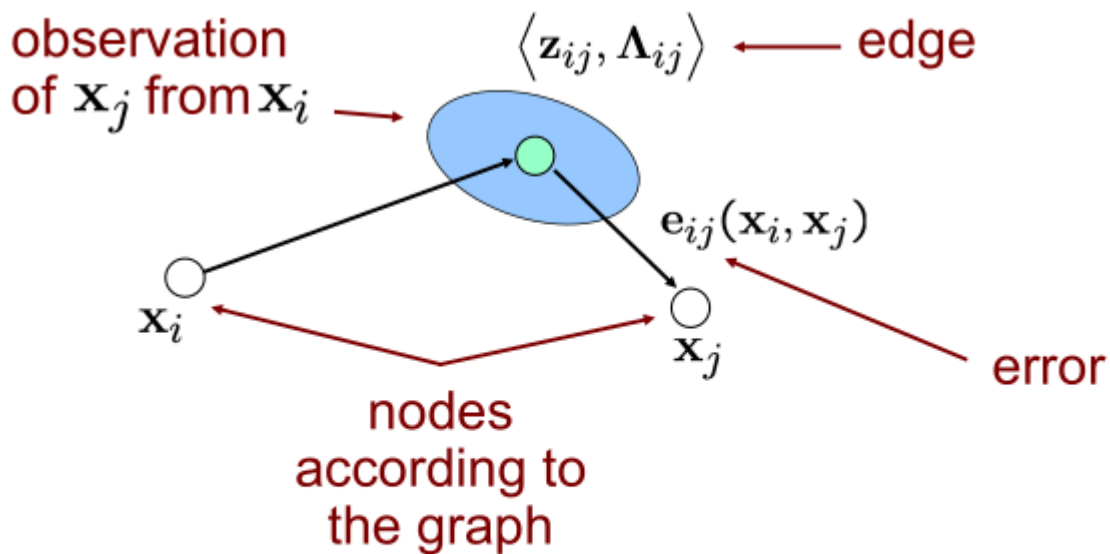
- Robust Kernels: Down-weight impact of outliers
 - Choice must align with outlier distribution
- Impacts Jacobians



◦

Graph-based SLAM

- **Node:** Robot Pose at given Point in Time
 - $x = x_{1:n}$
- **Edge:** Spatial constraint between 2 nodes
- **Graph-based SLAM**
 - **Graph Construction:** Build graph and
 - **Graph-Optimisation:** Find node configuration:
 - minimise error introduced by the constraints



Goal:
$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \sum_{ij} \mathbf{e}_{ij}^\top \Lambda_{ij} \mathbf{e}_{ij}$$

-
- Information matrix Λ_{ij} to encode uncertainty
 - The bigger, the more impact

1. Define error function

- Error Function for Least Squares estimation

$$x^* = \arg \min_x \sum_k e_k^T(x) \Lambda_k e_k(x)$$

2. Linearise error function

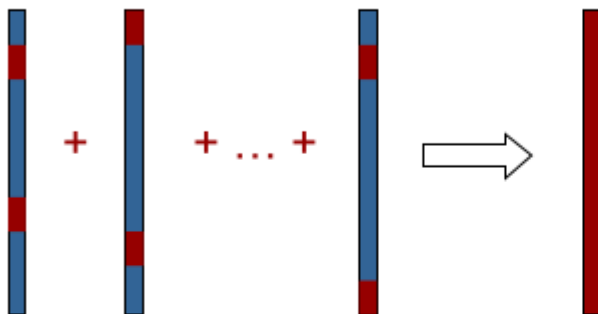
$$\mathbf{J}_{ij} = \begin{pmatrix} \mathbf{0} \dots \mathbf{0} & \underbrace{\frac{\partial \mathbf{e}(\mathbf{x}_i)}{\partial \mathbf{x}_i}}_{\mathbf{A}_{ij} \text{ } 3 \times 3} & \mathbf{0} \dots \mathbf{0} & \underbrace{\frac{\partial \mathbf{e}(\mathbf{x}_j)}{\partial \mathbf{x}_j}}_{\mathbf{B}_{ij} \text{ } 3 \times 3} & \mathbf{0} \dots \mathbf{0} \end{pmatrix}$$

◦

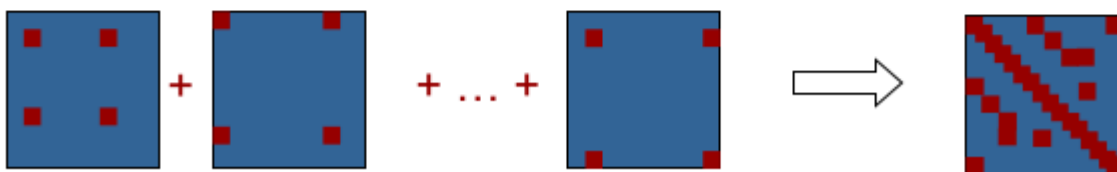
$$\mathbf{b}^T = \sum_{ij} \mathbf{b}_{ij}^T = \sum_{ij} e_{ij}^T \Lambda_{ij} \mathbf{J}_{ij}$$

$$\mathbf{H} = \sum_{ij} \mathbf{H}_{ij} = \sum_{ij} \mathbf{J}_{ij}^T \Lambda_{ij} \mathbf{J}_{ij}$$

$$\mathbf{b} = \sum_{ij} \mathbf{b}_{ij}$$



$$\mathbf{H} = \sum_{ij} \mathbf{H}_{ij}$$



-
- $\mathbf{H}$ matrix for a single edge

$$\begin{aligned}
H_{ij} &= J_{ij}^T \Lambda_{ij} J_{ij} \\
&= \begin{bmatrix} \vdots \\ A_{ij}^T \\ \vdots \\ B_{ij}^T \\ \vdots \end{bmatrix} \Lambda_{ij} \begin{bmatrix} \dots & A_{ij} & \dots & B_{ij} & \dots \end{bmatrix} \\
&= \begin{bmatrix} A_{ij}^T \Lambda_{ij} A_{ij} & A_{ij}^T \Lambda_{ij} B_{ij} \\ B_{ij}^T \Lambda_{ij} A_{ij} & B_{ij}^T \Lambda_{ij} B_{ij} \end{bmatrix}
\end{aligned}$$

- H is sparse, but b is not
 - Efficient solvers, like Cholesky decomposition

3. Build linear system

- Vector of states increments:

$$\Delta x^T = [\Delta x_1^T \quad \Delta x_2^T \quad \dots \Delta x_n^T]$$

- Coefficient vector:

$$b^T = [\bar{b}_1^T \quad \bar{b}_2^T \quad \dots \bar{b}_n^T]$$

- Normal equation matrix:

$$H = \begin{bmatrix} \bar{H}^{11} & \bar{H}^{12} & \dots & \bar{H}^{1n} \\ \bar{H}^{21} & \bar{H}^{22} & \dots & \bar{H}^{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{H}^{n1} & \bar{H}^{n2} & \dots & \bar{H}^{nn} \end{bmatrix}$$

- Compute error
 - $e_{ij}(x) = t_2 v(Z_{ij}^{-1}(X_i^{-1} X_j))$
- Compute blocks of Jacobian
 - $A_{ij} = \frac{\partial e(x_i, x_j)}{\partial x_i}$
 - $B_{ij} = \frac{\partial e(x_i, x_j)}{\partial x_j}$
- Update coefficient vector:
 - $\bar{b}_i^T + = e_{ij}^T \Lambda_{ij} A_{ij}$
 - $\bar{b}_j^T + = e_{ij}^T \Lambda_{ij} B_{ij}$
- Update Normal Equation matrix:
 - $\bar{H}^{ii} + = A_{ij}^T \Lambda_{ij} A_{ij}$
 - $\bar{H}^{ij} + = A_{ij}^T \Lambda_{ij} B_{ij}$
 - $\bar{H}^{ji} + = B_{ij}^T \Lambda_{ij} A_{ij}$

- $\bar{H}^{jj} + = B_{ij}^T \Lambda_{ij} B_{ij}$

4. Solve linear system

5. Iterate procedure until convergence

- One node needs to be fixed:

- Constraint, that sets $dx_1 = 0$